VD detector protection

Existing hardware, inhibit paths and VD additions

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DEEP UNDERGROUND NEUTRINO EXPERIMENT

Fermilab

DPS arbitrates protection; SC operates, monitors and logs

DPS	**Detector Protection System.** Uses facility and detector conditions to allow or block actions; issues permits, inhibits and trips.
SC	**Slow controls, using Ignition.** Operates and monitors the detector; owns the data and event logs, including DPS decisions.
BMS	**Building Management System.** Controls building services. FFDG BMS and SDSTA BMS are separate systems.
Local interlocks	**Equipment-level protection.** Can trip without waiting for SC or the central DPS.

An operating request cannot override a protection trip.

DPS receives inputs from four system domains

FFDG BMS	API through the Batavia server at FNAL → SC → DPS.
SDSTA BMS	Interface under investigation: available signals and connection points.
Cryogenics	Selected process conditions. Each inhibit names the actual instrument or controller that supplies them.
Detector	SC, the detector subsystems, and **rack infrastructure**: racks, PDUs and rack sensors.

Example: low liquid level withdraws the HV permit. SC logs the condition, DPS decision and equipment state.

The FFDG route is the working integration basis. SDSTA retains personnel-safety responsibility. Facility and cryogenic systems retain control of their services.

VD uses SNMP control and dedicated protection I/O

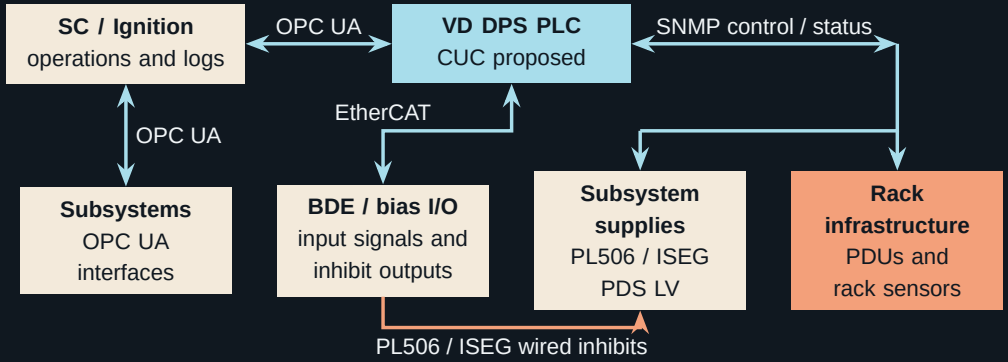


Diagram. PDUs and WIENER supplies share SNMP control/status. BDE/bias also have dedicated wired protection.

PLC = programmable logic controller; CUC = Central Utility Cavern; LV = low voltage. Additional subsystem exchanges go through SC. TDE has PDU control now; other interfaces remain options. HD may extend this system or use another PLC.

BDE already defines a distributed protection system

BDE = bottom drift electronics. I/O = inputs and outputs.

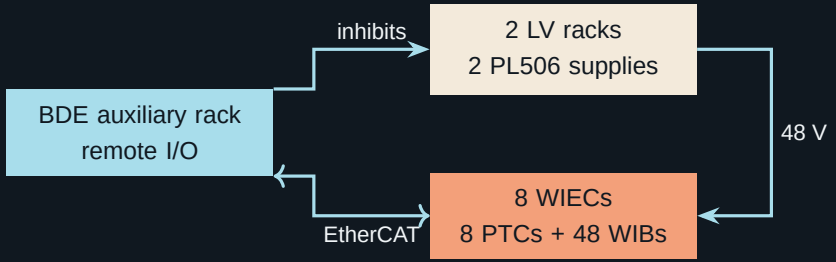


Diagram. Ten groups: 20 PL506 supplies, 80 WIECs and 480 WIBs.

The auxiliary I/O also connects fans, heaters and temperature sensors.

WIEC = warm interface electronics crate; PTC = power and timing card; WIB = warm interface board. June 2026 layout; fan/heater wiring omitted here.

BDE drawings, parts and software are already available

TPC protection panel designs

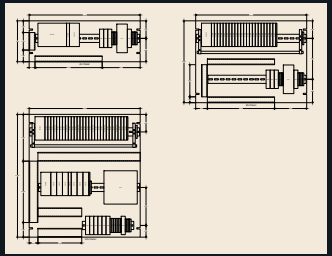
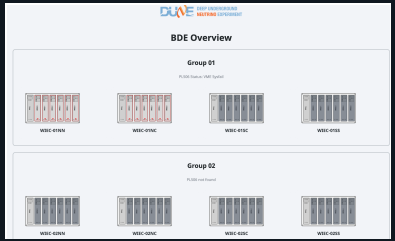


Diagram. Central PLC, bias I/O and LV/fan/heater reference panels.

BDE Ignition screens



Existing display of WIEC groups; June 2026 SC progress presentation.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

Historical TPC reference: EDMS 2401090 v4, sheet 2, with parts and power workbooks. The VD baseline is eight WIECs per group.

Trace each inhibit to the hardware it uses

The matrix gives the condition and action. Trace the full path for each affected load.

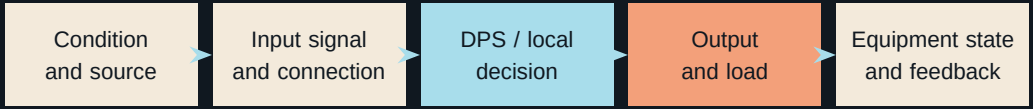


Diagram. Record the rack, connector, protocol, channel, power feed and allowed detection-to-shutdown time.

Use the existing rack lists, interfaces, panel drawings, parts and power calculations.

Result: hardware already covered, additions with quantities, and open choices that change those quantities.

Count shared inputs and outputs once, even when several inhibits use them. Wiring and software work are listed separately from hardware purchases.

BDE cooling already has an input-to-inhibit path

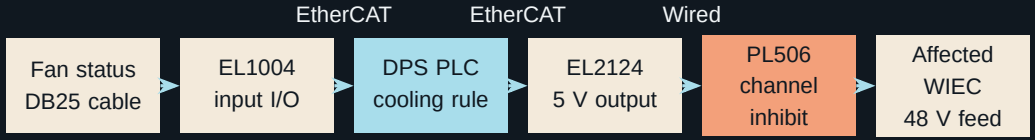


Diagram. Cooling fault → withdraw the permit for the affected PL506 channel.

Feedback: PL506 state and voltage/current → SNMP → DPS; OPC UA carries the result to SC for logging.

VD allocation check: six fan-status signals per WIEC means 48 per eight-WIEC group. Match these to the populated VD inputs and cables.

The six-fan WIEC note differs from the four-fan ICEBERG reference. These are baseline requirements, not new-module counts. Fan voting and allowed shutdown time remain to be agreed.

Rack infrastructure is a separate protection system

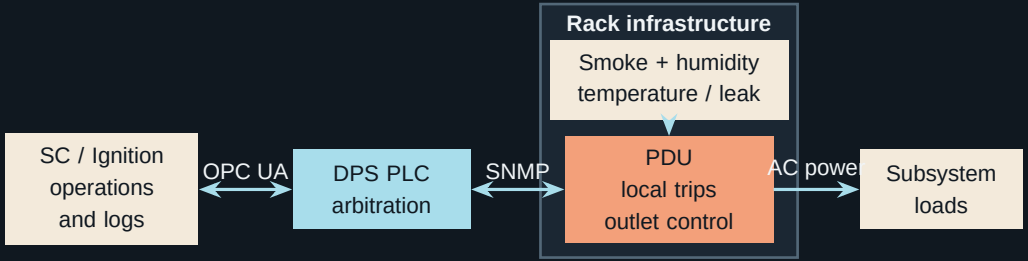


Diagram. Count the rack, PDUs, sensors and interface parts once. Link each subsystem to its PDU outlets.

Local PDU rules protect the mapped outlets. Humidity is monitored; a humidity shutdown needs a stated requirement.

The PX3 package supplies the smoke interface, powered contact hub and relay. Temperature/leak sensors are included where fitted. Reset and re-enable remain separate actions.

Detector additions follow the existing VD hardware

System	Starting hardware	What determines additions
BDE / HVS	BDE I/O; PL506 / ISEG inhibits; Heinzinger and cRIO	Signal and load mapping against the existing VD design
PDS / PoF	34 LV feeds; 80 PoF boxes	Opta PLC or Beckhoff PLC: local I/O, power, enclosures and cables
TDE	20 racks; current control through PDUs	Crate status / external inhibit retained as future options
CalCI	Existing instrumentation and calibration descriptions	Status, optical permits and actuator connections

The 80 PoF boxes have 1,280 monitoring values. Each controller option must cover the same physical signals and required actions.

PDS = photon detection system; PoF = power over fibre; TDE = top drift electronics; HVS = high-voltage system; CalCI = calibration and cryogenic instrumentation. These values are not a central DPS terminal count.

Facility interfaces need named signals and routes

Source	Hardware consequence of the path review
FFDG BMS	Use the API through Batavia and SC. Name the required points; the API exchange itself adds no discrete or analog DPS terminals.
SDSTA BMS	The survey identifies available signals, controllers and connection points. Those determine any interface hardware.
Cryogenics	Name the measurement or controller status and its transport. Some level measurements come through detector instrumentation and SC.

FAC-01 example: map the fire-alarm signal to the drift-HV and CRP/FC bias shutdown paths before assigning an additional facility input.

The known FFDG API route does not establish that every required alarm is exposed on it. SDSTA interface work remains under investigation.

Test arbitration, trips and recovery on the bench

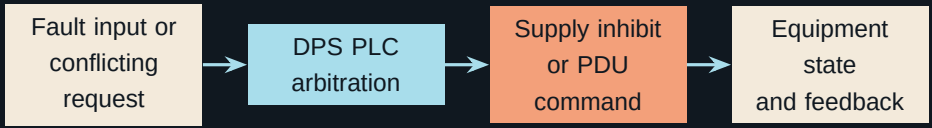


Diagram. Use the BDE vertical-slice work and G271 equipment for representative VD paths.

Measure detection through completed shutdown. Test local trips, SC and fieldbus loss, reset and re-enable. Verify that SC logs the decisions and equipment states.

No critical sub-ms function has been identified so far. Simulation prepares the fault and load cases; hardware tests measure the complete response.

Take the VD hardware to review in June 2027

Work before review	Result for procurement
Trace the VD inhibits	Existing hardware, added channels and cables, quantities and open alternatives
Build representative paths	Measured trip time, failure/recovery results and SC logging
Prepare purchases	Panel space and power, drawings, parts lists, costs, lead times and spares

After readiness review: order the hardware and install it in the site sequence.

Possible facility-interface need: summer 2027. Full VD response-matrix review: early 2028. The current path review establishes the requirements that affect hardware purchases.

I will lead the VD hardware work with a control engineer

Proposed contribution	Work
Manuel + control engineer	VD design, SURF interfaces and procurement package
Terri and Trevor	BDE panels, PLC/I/O implementation and bench work
Geoff + Manuel	SC commands, PDU protection and recovery
Vivian + consortia / DPS	Equipment requirements and ICD updates

Support needed: control-engineering time and access to the existing test stands.

SURF work: inspection maps are ready; access is being coordinated with Rachel, with Dale Curran identified as an SDSTA integration contact.

Backup: VD rack space and the existing DPS allocation

Rack type	Count	DPS allocation and workbook space
BDE auxiliary	10	12U DPS; U3–U21 marked Space (19U); eight WIECs per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U DPS; U3–U12 marked Space (10U)

Space marked in the rack workbook is not a new DPS reservation. Free DIN-rail length and I/O capacity depend on the populated VD panel.

May 2025 VD workbook: BDE Aux Rack and Bias HV Rack. Auxiliary U9 also has a possible ground-plane readout note. U is rack height.

The VD bias-panel draft specifies two DIN rails per rack

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

May 2025 workbook, Bias HV Rack!K17:K21. This is the first-pass layout; the difference from 576 bias channels is not verified spare capacity.

Use the component loads for the VD power budget

Equipment	Power / capacity	Basis and limit
PL506 input	2.25 kW each	20 supplies: about 45.1 kW expected AC input
WIEC fans only	380 W per group	$8 \times 6 \times 0.33\,\text{A} \times 24\,\text{V}$; excludes heaters and I/O
Bias DPS supply allowance	24 V, 40 A	960 W DC capacity per rack in the workbook

The inherited 831 W auxiliary figure needs a VD component budget covering fans, heaters and populated I/O.

May workbook; WIEC safety note, internal revision 0. Fan arithmetic assumes six fans per WIEC. Rated load, expected AC input and DC supply capacity are distinct.

Rack infrastructure already has a hardware reference

Rack component	Reference hardware / allocation
PDU	Raritan PX3; inlet and outlet allocation for the connected loads
Smoke detector	4WTAR-B and EOLR-1 end-of-line relay
Contact interface	DX-PD2C5; 12 V supply to the smoke detector
Humidity	Record the probe and peripheral interface for each rack
Temperature / water leak	PDU peripheral sensors where fitted

The May workbook includes a 76-entry rack-protection/PDU allowance. Map the actual rack packages against this existing allowance.

DUNE-doc-32655 v8, sections 3 and 5; May workbook, Standard Rack and Power Summary!B28. The guide does not specify a humidity probe model or trip requirement.

Reference: local PDU trips and recovery

PX3 setup guide	Local behaviour documented
Smoke, water, temperature	Local event rules and scripts can switch off the protected outlets
Network-switch outlet	Kept powered during the guide's fallback shutdown
After a local trip	Power is restored deliberately; no automatic restoration

The VD control chain is Ignition → DPS PLC → PDU. Local trip status returns to the PLC and to SC for logging.

DUNE-doc-32655 v8 describes an Ignition-direct prototype with local fallback on loss of Ignition. The VD PLC command path requires matching reachability, trip and reset configuration; it is not a claim that the prototype already implements it.

The VD inventory covers the other subsystem connections

Subsystem	Working inventory and interfaces
PDS readout / calibration	34 LV feeds in the current presentation scope; supply status and control interfaces
Power over fibre (PoF)	80 boxes; 640 temperature and 640 voltage readings; Opta PLC or Beckhoff PLC alternative
TDE	20 mezzanine racks; PDU control now; other TDE interfaces are possible extensions
Calibration / cryogenic instrumentation	Level and temperature readout, purity monitors, ionization laser and neutron source

Additional subsystem exchanges use OPC UA through SC. June 2026 SC Progress, slide 29, names the PoF controller alternatives. Box count and monitoring values do not establish controller or central DPS terminal quantities.

The BMS drawing identifies 20 control-panel symbols

Drawing area	Interfaces on the inspection maps
CUC, level 4850	Eight panel references: air handling, chilled water, power and associated equipment
Level 4910	North and south sump interfaces, SP-1 and SP-2
4850-08-ES group	Spare panel for future equipment; mapped to the access-drift inspection area
Other 4850 areas	Chillers, pumps, ventilation and generator interfaces

Inspection planned with Rachel; Dale Curran is the SDSTA integration contact identified for this work.

U-FD-M-901 rev12, 14 February 2025; marked 4850/4910 maps and survey register. Symbols are drawing references; field observations are pending.

Reference: the May 2025 VD rack inventory

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

May 2025 allocation: rack positions, distinct from PLC panels and remote-I/O stations. Includes the listed reserved and spare positions.